DRUG DISCOVERY · BUILD FELLOWSHIP

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Targeting P. falciparum

Research scope: This Build Fellowship deliverable presents a literature-informed, proposed PfTMK drug-discovery workflow. Unless specifically accompanied by original laboratory records and results, the assays, quality criteria and downstream experiments described are planned approaches, not experiments performed or validated by the author.

Thymidylate Kinase (PfTMK)

Exploring the drug development cycle and strategic approaches to anti-malarial treatment

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Weeks	Techniques	Pipeline Stages	Novel Target

WEEK 1 DELIVERABLE | Introduction to Malaria and PfTMK

MALARIA & PfTMK: THE CASE FOR A NEW TARGET

Malaria: The Global Challenge

200M+	500K	~40%
Infections / yr	Deaths / yr	Seq. similarity to huTMK

- P. falciparum is the deadliest species, with the highest mortality and severe clinical cases worldwide.
- Rising ACT resistance demands urgent discovery of new parasite-specific drug targets.
- New targets must exploit pathways structurally distinct from the human host.

PfTMK: An Essential Molecular Target

Core Function	Catalyzes phosphorylation of dTMP→dTDP, a vital step in producing the DNA precursor dTTP.
Metabolic Bottleneck	Parasite relies on de novo pyrimidine synthesis during blood stage a critical survival checkpoint.
High Selectivity	Only ~40–41% similarity with human TMK, enabling design of parasite-specific inhibitors.
Substrate Plasticity	Unlike human TMK, PfTMK uses dGMP as substrate a unique structural opening for drug design.

WEEK 2 DELIVERABLE | Expression System · Affinity Tag · QC Measures

EXPRESSION, PURIFICATION & QUALITY CONTROL

Expression System, Tag & Purification

Host — E. coli	High yield, fast growth, compatible with $^{15}N/^{13}C$ isotopic labelling for NMR.
Chaperones — GroEL/ES	Co-expression ensures correct PfTMK homodimer folding; prevents inactive inclusion bodies.
Tag — N-terminal 6×His	Selective IMAC binding via Ni-NTA/Cobalt resin; small footprint preserves dimer interface.

Three-Step Purification Pipeline

01 Cell Lysis	Sonication releases soluble PfTMK from E. coli host cells.
02 IMAC Affinity	Ni-NTA resin captures 6×His-tagged PfTMK selectively, removing the bulk of host-cell proteins.
03 SEC Polishing	Confirms homodimer form and removes aggregates, ensuring >95% homogeneity before HTS.

Quality Control Measures

Analytical SEC

Analytical size-exclusion chromatography (SEC) can help characterize oligomeric state and detect aggregates before screening; enzyme activity should be independently verified. Critical because dGMP binding induces dynamics at the dimer interface that cannot be measured in aggregated samples.

SDS-PAGE Gel Electrophoresis

Visual confirmation of protein purity and correct monomer size (~24 kDa). Helps assess purity and detect substantial host-cell protein contamination, but cannot establish complete absence of contaminants that would interfere with NADH-coupled kinase assay signals. A proposed purity acceptance criterion of >95% would require justification and analytical verification before screening.

WEEK 3 DELIVERABLE | Hit Identification Methodology & Rationale

HIT ID: NADH-COUPLED HTS ASSAY

Enzymatic Cascade & Strategic Rationale

Continuous spectrophotometric assay — ↓ absorbance at 340 nm = real-time PfTMK activity.

Reaction 1	PfTMK phosphorylates dTMP → dTDP (ATP consumed)
Reaction 2	Pyruvate Kinase: ADP + PEP → ATP + Pyruvate
Reaction 3	LDH: Pyruvate + NADH → Lactate + NAD^+ (↓ 340 nm)

- Exploits ~40% PfTMK/huTMK sequence divergence for parasite selectivity.
- Real-time format distinguishes competitive vs. non-competitive hits early.
- Appropriate for focused library (<5,000 compounds) manageable reagent window.

⚠ **Note:** NADH degrades over time, semi-HTS only. ADP-Glo used for larger screens.

Week 3 Deliverable: Why This Method?

✓ WHY NADH-COUPLED ACTIVITY ASSAY	✕ WHY NOT THE OTHERS
✓ **Directly** measures enzymatic activity not a proxy or endpoint signal. ✓ **Real**-time kinetics identifies inhibition mechanism from the outset. ✓ **Focused** PfTMK library (<5,000 cpds) keeps NADH degradation manageable. ✓ **Acknowledged** limitation: NADH instability limits throughput to semi-HTS; ADP-Glo preferred for libraries >10,000 compounds.	✕ **Binding** Assay HTS: Detects binding only cannot confirm enzymatic inhibition. ✕ **Cell**-Based HTS: Too early off-target effects confound target inhibition.

WEEK 4 DELIVERABLE | H2L Biophysical Technique

BIOPHYSICAL VALIDATION: SPR & RATIONALE

Surface Plasmon Resonance (SPR)

- Label-free: measures real binding without perturbing compound structure.
- Full kinetics: kon, koff and KD for rigorous hit prioritisation.
- Binding residence time (slow dissociation rate) can contribute to hit prioritization, but does not independently predict in vivo efficacy.
- Selectivity Index: parallel human TMK measurements could assess differential binding; functional selectivity would require additional biochemical and cellular testing.
- Requires only micrograms of protein compatible with limited reagent supply.
- MST used as orthogonal in-solution confirmation (4 µL sample; no chip needed).

Week 4 Deliverable: Chosen Method

CHOSEN: Surface Plasmon Resonance (SPR) ✓ WHY THIS METHOD	✕ WHY NOT THE OTHERS
✓ **Label**-free full kinetics (kon/koff/KD) for rigorous ranking. ✓ **Residence** time data supports hit prioritization but does not independently establish in vivo efficacy. ✓ **Parallel** huTMK chip gives Selectivity Index in same run.	✕ **ITC**: Requires milligrams of protein; very low throughput. ✕ **DSF**: Thermal stability shifts only no affinity or kinetics. ✕ **CD**: Reports secondary structure, not ligand binding events.

WEEK 5 DELIVERABLE | H2L Biochemical Technique

BIOCHEMICAL CHARACTERISATION & RATIONALE

NADH-Coupled Kinase Assay (Primary) + ADP-Glo (Orthogonal)

- PfTMK is a kinase that directly measures phosphoryl-transfer activity in real time.
- Continuous format: full IC50 curves & inhibition mode determination.
- Sensitive to ATP/ADP turnover precisely matched to dTMP phosphorylation.
- Parallel human TMK run provides direct selectivity ratio per compound.
- ADP-Glo (luminescent, end-point) used as orthogonal confirmation detects low-potency hits with minimal background interference from coloured compounds.

Week 5 Deliverable: Chosen Method

CHOSEN: NADH-Coupled Continuous Kinase Assay ✓ WHY THIS METHOD	✕ WHY NOT THE OTHERS
✓ **Directly** measures kinase activity not a proxy or endpoint signal. ✓ **Continuous** readout enables IC50 and mechanism-of-inhibition determination. ✓ **Scalable** to 384-well plates for cost-effective H2L profiling.	✕ **TR**-FRET: Antibody reagents specific to dTDP product not available. ✕ **ADP**-Glo: End-point cannot determine inhibition mechanism alone. ✕ **AlphaLISA**: Proximity-based; prone to compound colour interference.

WEEK 6 DELIVERABLE | Resolving PfTMK-Inhibitor Complexes for SAR

STRUCTURAL BIOLOGY & RATIONALE

X-ray Crystallography (Co-crystal Structures)

- Atomic resolution (<2.0 Å): maps H-bonds and van der Waals contacts at binding site.the
- SAR Optimisation: identifies functional groups driving potency for rational modification.
- Crystal soaking rapidly validates multiple analogues once the protocol is established.
- Side-by-side PfTMK vs. huTMK structures reveal parasite-specific pockets for selectivity.
- PfTMK has been crystallised; a previously established protocol reduces time to structure.
- Cryo-EM considered for any larger PfTMK-protein complexes identified downstream.

Week 6 Deliverable: Chosen Method

CHOSEN: X-ray Crystallography ✓ WHY THIS METHOD	✕ WHY NOT THE OTHERS
✓ **Atomic** resolution (<2.0 Å) for precise binding-site mapping. ✓ **Crystal** soaking enables rapid analogue testing for SAR. ✓ **PfTMK** crystallised before reduces time to first structure.	✕ **Cryo**-EM: PfTMK homodimer (~48 kDa) too small for high-res cryo-EM. ✕ **NMR**: Lower resolution; size limit; requires isotopic labelling. ✕ **HDX**-MS: Solvent accessibility only insufficient for drug design.

WEEK 7 DELIVERABLE | Lead Optimisation — Cellular Target Engagement

CELL-BASED TECHNIQUE & RATIONALE

NanoBRET™ Target Engagement Assay

- Measures direct compound-target engagement inside living cells physiologically relevant.
- PfTMK tagged with NanoLuc luciferase; fluorescent tracer competes with test compound.
- BRET signal decreases when compound displaces tracer confirms cellular engagement.
- Accounts for cell permeability, metabolic stability, and efflux pump effects.
- Cellular IC50 bridges biochemical potency to in vivo efficacy prediction.
- Orthogonal to NADH biochemical assay together form a robust hit-to-lead filter.

Chosen Method

CHOSEN: NanoBRET™ Target Engagement Assay ✓ WHY THIS METHOD	✕ WHY NOT THE OTHERS
✓ **Confirms** PfTMK binding inside living cells not just in vitro.	✕ **Reporter** Gene: Downstream transcriptional response too indirect for kinase.
✓ **Directly** measures cell permeability alongside target engagement.	✕ **FACS**: Cell health endpoints only cannot confirm PfTMK engagement.
✓ **Low** false-positive rate only permeable stable compounds signal.	✕ **SYBR** Green: Measures growth inhibition only not mechanism of action.

WEEK 7 DELIVERABLE | Preclinical In Vivo Proof-of-Concept

ANIMAL MODEL & RATIONALE

P. berghei Mouse Model (Peters 4-Day Suppressive Test)

- Gold standard rodent malaria model: P. berghei is the closest relative of P. falciparum in mice.
- Peters 4-Day Test: compound dosed for 4 days; parasitaemia quantified by blood smear.
- Can support evaluation of antimalarial activity and tolerability; pharmacokinetics require separately planned sampling and analysis.
- Rapid turnaround (~5 days) enables iterative SAR-driven optimisation cycles.
- Regulatory precedent: widely accepted by FDA/EMA as primary antimalarial proof-of-concept.
- Can provide early preclinical evidence, but is not sufficient by itself to justify human clinical candidate selection.

Week 7 Deliverable: Chosen Method

CHOSEN: P. berghei Mouse Model ✓ WHY THIS METHOD	✕ WHY NOT THE OTHERS
✓ **Closest** P. falciparum relative in rodents — high translational value. ✓ **4**-Day test measures efficacy, PK and tolerability simultaneously. ✓ **Accepted** by FDA/EMA as primary anti-malarial proof-of-concept.	✕ **Humanised** Mouse: Requires human RBC engraftment — complex and expensive. ✕ **NHP** Model: Too resource-intensive for early lead optimisation stage. ✕ **Zebrafish**: No P. falciparum infection; poor PK/PD translation.

WEEK 8 DELIVERABLE · BUILD FELLOWSHIP

Conclusion: A Robust Pipeline for Anti-Malarial Drug Discovery

01 Target Validation	PfTMK was evaluated as a potential antimalarial drug-discovery target due to its essential role in the parasite's de novo pyrimidine synthesis and unique substrate plasticity with dGMP.
02 Assay Robustness	The proposed screening funnel combines functional assays with SPR binding kinetics to inform hit prioritization; no experimental hit ranking is established here.
03 Selectivity Strategy	Parallel human TMK assays and structural analysis were proposed to investigate selectivity. No selectivity window or host-toxicity reduction was demonstrated in this project.
04 Path to Clinic	Cellular target-engagement studies and an appropriate preclinical efficacy model are proposed next steps, not completed regulatory proof-of-concept studies.

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